

Partial Result Packet

A two-cycle obstruction to bounded geodesic repair

Status: partial result, likely valid

Source problem and scope

The source is Cécilia Lancien, *High dimension and symmetries in quantum information theory*, arXiv:1607.06843. A local copy is `source_paper.pdf`. For permutations on p points, write

$$d(\sigma, \tau) = p - \#(\sigma^{-1}\tau)$$

for the transposition Cayley distance. The source asks whether excess 2δ over a geodesic can be repaired within distance $O(\delta)$, with polynomially bounded fibres.

Remark 9.16.5. *The upper bound we established in Lemma 9.16.4 is probably far from optimal (e.g. it is likely that the exponent 4δ in the polynomial pre-factor in k and p can be improved). But this does not really matter for our specific goal. Nonetheless, in the special case of non-geodesic permutations between id and γ on $\{1, \dots, p\}$, it is in fact quite easy to obtain an upper bound which scales as $p^{3\delta}$ for the ratio between the number of 2δ non-geodesic permutations with a given number of cycles and the number of geodesic permutations with the same number of cycles. We present the result in Lemma 9.16.6 below, the problem being that the proof method does not seem to generalize so straightforwardly to the case that we truly need, that is the one of non-geodesic permutations between id and γ_f .*

Note also that very similar looking upper bounds had previously been derived regarding the cardinality of the set $\mathfrak{S}_\delta(p) = \{\alpha \in \mathfrak{S}(p) : \#(\gamma^{-1}\alpha) + \#(\alpha) = p + 1 - 2\delta\}$, which is the union of the sets $\mathfrak{S}_{\delta,m}(p)$ defined in Lemma 9.16.6 for $1 \leq m \leq p$. In particular, it was established in [142], Lemma 12, that for any $0 \leq \delta \leq \lfloor p/2 \rfloor$,

$$|\mathfrak{S}_\delta(p)| \leq |\mathfrak{S}_0(p)| p^{3\delta} = \text{Cat}_p p^{3\delta}.$$

However, this is definitely even less enough for our purpose: the latter really requires an upper bound on the number of permutations which have a given defect and a given number of cycles in terms of the number of permutations which have no defect and the same (or a related) number of cycles.

On the other hand, one may have hoped for a stronger result than these simply counting ones. For instance something like

$$d(\text{id}, \alpha) + d(\alpha, \gamma) = d(\text{id}, \gamma) + 2\delta \Rightarrow \exists \alpha' : d(\alpha, \alpha') = 2\delta' \text{ and } d(\text{id}, \alpha') + d(\alpha', \gamma) = d(\text{id}, \gamma),$$

with $\delta' \leq \theta\delta$ and with the mapping $\phi : \alpha \mapsto \alpha'$ satisfying $|\phi^{-1}(\alpha')| \leq p^{\kappa\delta}$, for some coefficients θ, κ . However, determining whether this kind of statement holds or not seems to remain an open question.

Lemma 9.16.6. *Let $p \in \mathbf{N}$ and denote by γ the canonical full cycle on $\{1, \dots, p\}$. For any $0 \leq \delta \leq \lfloor p/2 \rfloor$ and*

The displayed question is for one canonical full cycle γ . The preceding paragraph says that the case truly needed in the application is γ_f , the product of canonical cycles on the level sets of f . This packet disproves that multi-cycle extension with two level sets. A sibling counterexample packet produced later in the run settles the printed single-full-cycle question negatively.

Proof intuition

Take two cyclic blocks A and B . Let γ_f rotate each block, while α pairs points across the blocks in reverse cyclic order. Both α and $\alpha^{-1}\gamma_f$ are perfect matchings, so the defect is one. However, every permutation on a shortest path from id to γ_f preserves both blocks, while α swaps them. The quotient must alternate between the blocks, giving linear distance.

Interval lemma

Lemma. *If $d(\text{id}, \pi) + d(\pi, \gamma) = d(\text{id}, \gamma)$, then every cycle of π is contained in a cycle of γ .*

Proof. Concatenate shortest paths from id to π and from π to γ . The equality makes this a shortest path. At every step from the identity, the transposition must increase length by one, hence merge two cycles. A split would decrease length. Thus cycle partitions only become coarser, proving the claim. $\square$

Main result

Theorem. *For every $n \geq 2$, there are $\gamma, \alpha \in S_{2n}$ such that γ is a product of two n -cycles,*

$$d(\text{id}, \alpha) + d(\alpha, \gamma) = d(\text{id}, \gamma) + 2,$$

and

$$d(\alpha, [\text{id}, \gamma]) = n.$$

Proof. Use subscripts in $\mathbb{Z}/n\mathbb{Z}$. On

$$A = \{a_i : i \in \mathbb{Z}/n\mathbb{Z}\}, \quad B = \{b_i : i \in \mathbb{Z}/n\mathbb{Z}\},$$

define

$$\gamma = (a_0 \ a_1 \ \cdots \ a_{n-1})(b_0 \ b_1 \ \cdots \ b_{n-1}), \quad \alpha = \prod_i (a_i \ b_{-i}).$$

The permutation α is a fixed-point-free involution, so $\#(\alpha) = n$. Moreover

$$\alpha\gamma(a_i) = b_{-i-1}, \quad \alpha\gamma(b_j) = a_{-j-1}.$$

Hence $\alpha^{-1}\gamma = \alpha\gamma$ is another fixed-point-free involution and $\#(\alpha^{-1}\gamma) = n$. Therefore

$$d(\text{id}, \alpha) = n, \quad d(\alpha, \gamma) = n, \quad d(\text{id}, \gamma) = 2n - 2,$$

which proves the excess-two identity.

If $\pi \in [\text{id}, \gamma]$, the interval lemma says that π preserves A and B . Since $\alpha^{-1} = \alpha$ swaps them, $\alpha^{-1}\pi$ also swaps them. Every cycle of this quotient alternates between A and B , so it has at most n cycles. Consequently

$$d(\alpha, \pi) = 2n - \#(\alpha^{-1}\pi) \geq n.$$

Equality holds at $\pi = \text{id}$, proving the claimed exact distance. $\square$

Corollary. *No universal $\theta < \infty$ gives a repair bound $d(\alpha, \alpha') \leq 2\theta\delta$ for every product γ_f of canonical cycles.*

Proof. The source's multi-cycle defect equation is

$$\#(\gamma^{-1}\alpha) + \#(\alpha) = p + \#(\gamma) - 2\delta.$$

Here the left side is $2n$, while $p = 2n$ and $\#(\gamma) = 2$, so $\delta = 1$. The required distance is nevertheless at least n , which is unbounded. The additional fibre condition therefore cannot rescue the claim. $\square$

Verification, limitations, and novelty

The proof is uniform and computationally independent. The script `code/verify_obstruction.py` exhaustively enumerates S_{2n} for $2 \leq n \leq 4$, selects the interval by its distance equality, and returns:

```
n=2: interval_size=4, excess=2, min_distance=2
n=3: interval_size=25, excess=2, min_distance=3
n=4: interval_size=196, excess=2, min_distance=4
```

The result in this packet is partial because its block obstruction uses two cycles. Subsequent work in the same run found a different defect-one family for a single full cycle whose distance from the geodesic interval is unbounded; see the sibling full-cycle counterexample packet in this run. Thus the exact printed question is now also answered negatively.

A bounded search of the run indexes and arXiv for the source identifier, permutation genus, absolute order, geodesic-interval distance, and two-cycle products found related work on noncrossing and plane permutations, but no occurrence of this explicit defect-one obstruction. Standard genus-slicing operations may move the face cycle and do not supply the fixed-face statement.

Human review focus. Check the source convention $\#(\gamma_f^{-1}\alpha) + \#(\alpha) = p + \#(\gamma_f) - 2\delta$ and the interval lemma. These two points make the scope and the defect computation transparent.

References

- C. Lancien, *High dimension and symmetries in quantum information theory*, arXiv:1607.06843, Section 9.16.3.
- R. X. F. Chen and C. M. Reidys, *Plane permutations and applications to a result of Zagier–Stanley and distances of permutations*, arXiv:1502.07674.
- R. Cori and G. Hetyei, *Counting genus one partitions and permutations*, arXiv:1306.4628.
- A. Hock, *Genus permutations and genus partitions*, arXiv:2306.16237.